

Ranadeep Biswas, Ph.D.

Informal Systems

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github.com/rnbguy

Experience

• Informal Systems

- Model-based Testing using Model Checking.
 - July 2021 - December 2022
 - **Modelator**: Model-based testing framework using TLA+ and Apalache.
 - **Atomkraft**: E2E Testing framework for Cosmos-SDK blockchains.
- Software auditor for blockchain technologies.
 - January 2023 - June 2023
 - **Blockchain Audits**:
 - Gravity Bridge
 - IBC or Protocol
 - Osmosis
 - Mars Protocol
 - Namada
 - Axelar
- Developing and maintaining IBC in Rust
 - July 2023 - June 2024
 - **ibc-rs**: IBC in Rust
- Developing IBC for Starknet
 - July 2024 - Now
 - **ibc-starknet**: IBC in Cairo as Starknet contract.

Interests

- Blockchain and interoperability
- Consensus and distributed systems
- Formal methods and verifiable AI
- Open-source contribution

Education

• Ph.D. in Computer Science

- University of Paris with IRIF
- November 2017 - March 2021
- Thesis: *Testing Weakly Consistent Distributed Storage Systems*
- Publications available on my website.

• Master of Science in Computer Science

- Chennai Mathematical Institute
- August 2015 - April 2017
- Got introduced to Software Verification via CBMC.

• Bachelor of Science in Mathematics and Computer Science

- Chennai Mathematical Institute
- August 2012 - April 2015